

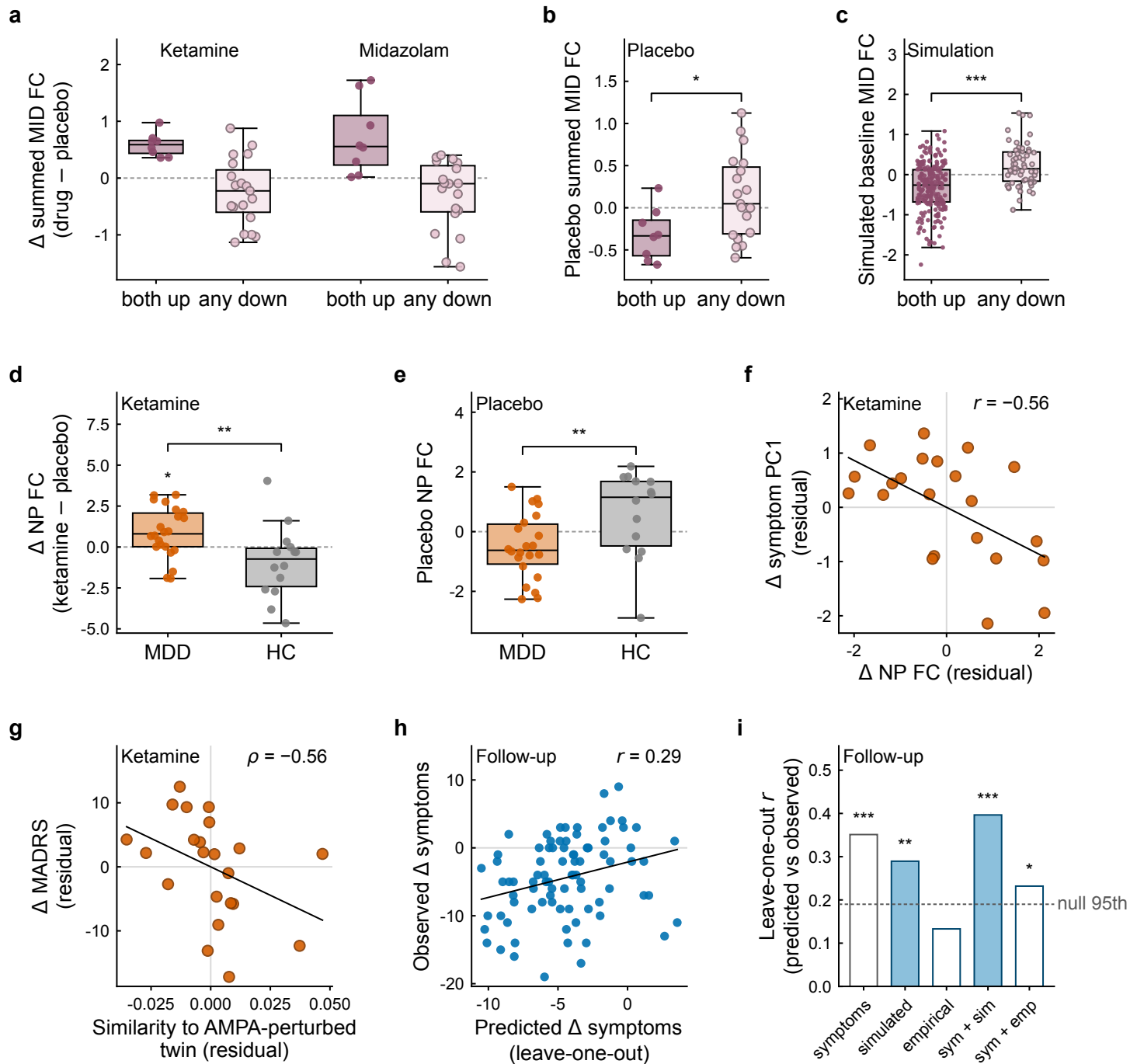


Fig. 5 | In vivo pharmacology and longitudinal association. Two cohorts test the perturbation account in vivo, one panel gives its model counterpart, and a third cohort tests it prospectively. Healthy volunteers ($n = 27$) received placebo, ketamine and midazolam; the clinical cohort ($n = 36$) received placebo and ketamine; the digital twins in c are the Fig. 4 cohort ($n = 288$); the longitudinal cohort is the IMAGEN follow-up ($n = 85$). Box plots show the median, 25th–75th percentiles and $1.5 \times$ IQR whiskers with every participant overplotted; dashed line, no change. An asterisk above a single box (d) is a two-sided one-sample t test against zero; brackets are the between-group test named below. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. The heading inside each panel names the condition or data source it shows. b, c and e make one statement in the model: the group whose connectivity rises under the drug is the group that starts from the lower drug-free level. In f, g and h both axes are residuals, so the grey cross marks zero on each and the coefficient in the heading band is the effect size itself (both axes being residuals, it is a partial coefficient); its P , confidence interval and n are given below. **a**, change in summed MID FC from placebo in the healthy cohort, split by the Fig. 4 response level (both up $n = 8$, any down $n = 19$). Ketamine: $+0.59 \pm 0.21$ in both-up versus -0.25 ± 0.58 in any-down; midazolam: $+0.72 \pm 0.66$ versus -0.30 ± 0.61 (mean $\pm$ s.d.). This panel is descriptive and carries no test: the two groups are defined by the sign of the very quantity plotted here, so any test of it would be circular. **b**, drug-free (placebo) level of the same two groups: -0.32 versus $+0.13$, two-sided Welch $t = -2.83$, $P = 0.01$. The grouping is defined on the drug-induced change and the plotted quantity is the drug-free level, so this contrast is not circular. **c**, the same comparison in the digital twins of the Fig. 4 cohort: simulated baseline MID FC under the both-up / any-down split defined on the NP sum (both up $n = 229$, any down $n = 59$): -0.31 ± 0.58 versus $+0.22 \pm 0.51$, Welch $t = -6.95$, $P < 0.0001$ – the model reproduces the direction seen in b on the same measure. **d**, ketamine-induced change in NP FC in the clinical cohort (MDD $n = 22$, HC $n = 14$): $+0.87$ versus -0.93 . The bracket is the group coefficient of an OLS model adjusted for age, sex, infusion order and placebo-session mean framewise displacement (beta = 2.43, $t = 3.43$, $df = 30$, $P = 0.0018$), not a raw two-sample test. **e**, placebo NP FC in the same cohort, same adjusted model (beta = -1.34, $t = -2.77$, $df = 30$, $P = 0.0096$). **f**, change in NP FC against change in the symptom PC1 in the patients ($n = 22$); both axes are residuals on baseline severity, age, infusion order and placebo-session mean framewise displacement, so the residual df is 16: partial $r = -0.56$, $P = 0.016$. **g**, similarity of the ketamine state to the AMPA-perturbed digital twin against MADRS change ($n = 22$), both residualised on age, sex, infusion order and placebo-session mean framewise displacement: Spearman $\rho = -0.56$, $P = 0.0069$; permutation $P = 0.0068$. **h**, leave-one-out predicted against observed four-year symptom change from the 12 NP edges of each participant's unperturbed digital twin ($n = 85$): $r = 0.29$, $P = 0.0072$; cross-validated $R^2 = 0.012$. Negative values on both axes are improvement; the line is a least-squares fit shown to indicate direction, not an additional test. **i**, leave-one-out correlation between predicted and observed symptom change for five model specifications in the same $n = 85$: age-19 symptoms alone $r = 0.35$, the 12 simulated edges $r = 0.29$, the 12 empirical edges $r = 0.13$, and symptoms combined with each edge set $r = 0.40$ and $r = 0.23$. Filled bars, simulated network; open bars, empirical network. The dashed line is the 95th percentile (0.19) of a permutation null for the 12-edge simulated model in which the outcome was shuffled and the whole leave-one-out procedure repeated 2,000 times (observed $P = 0.0095$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ on the leave-one-out correlation itself.